

EXPERIENCE

Software Engineer III | Meta

October 2023 – Present (Contract)

- Engineered Contextual Artificial Intelligence (AI) research prototypes for Reality Labs Research, using **Python**, **JavaScript**, **TypeScript**, **Kotlin**, **React**, **GraphQL** and **C#** to develop AI-driven experiences shaped by real-time environment data and user input.
- Implemented multimodal data pipelines and data annotation tools in collaboration with research scientists, optimized data streaming for AI inference in **RAG models**, deployed using **Docker** and **Kubernetes**
- Created a **Thrift** API to manage **LLM** prompt templates, ensuring structured validation and processing of results
- Led UX development for a multimodal contextual AI demo, showcased to the CTO and other senior leadership, influencing AI integration strategies for upcoming Meta wearable device research

Research Engineer | Meta

September 2022 – December 2022 (Internship)

- Built a multimodal **Augmented Reality (AR)** prototype for **HoloLens** using **Unity 3D (C#)**, **Python**, and **C++**
- Explored novel use cases for augmented reality, focusing on multimodal interactions, and presented key findings to researchers

Research Intern | Roblox

May 2022 – August 2022 (Internship)

- Developed a multiplayer experience on **Roblox Studio** using **Luau** that could be accessed via desktop or **Virtual Reality (VR)**
- Mentored a research assistant and [published](#) research on the effects of customizability on creativity in user-generated content

Research Intern | Adobe

May 2021 – December 2021 (Internship)

- Developed a typography layout prototype using **TypeScript** and an online experiment built using **JsPsych** and **Firebase**
- [Authored](#) peer-reviewed research and secured a [patent](#) on text layout based on semantic importance

Research Assistant | Dynamic Graphics Project, University of Toronto

July 2020 – August 2020 (Contract)

- Prototyped a novel tool for VR design space visualization and exploration in **Unity 3D (C#)**, and [published](#) research findings

Software Developer | Autodesk

May 2019 – August 2019 (Internship)

- Developed features for **Autodesk Maya's** Render Setup using **PyMEL**, **Python** and **Qt**, improving rendering workflows for 3D artists
- Implemented the [grouping feature](#) in Maya 2020, enabling organization and manipulation of operations during render setup

Software Developer | Intel

May 2018 – May 2019 (Internship)

- Designed Python and PostgreSQL infrastructure, streamlining product information management across international teams
- Extended and maintained a set of automated dashboards for project completion statistics and tracking
- Used mathematical models to predict FPGA static power consumption and wrote supporting software in **C++** and **Python**

Software Designer | Nokia

May 2017 – August 2017 (Internship)

- Automated daily build testing, increasing efficiency by 16x through **bash scripting** and process consolidation

EDUCATION

Master of Mathematics (MMath Thesis), Computer Science

University of Waterloo

- Research Areas: **Human-Computer Interaction**, **Virtual Reality**, **Spatial Augmented Reality**, **Creativity Support Tools**
- Thesis: [Perceptual Allowances of Anamorphic Interaction Cues in Spatial Augmented Reality](#)

Honours Bachelor of Science (HBS), Computer Science

University of Toronto

- Graduated with **High Distinction** and a specialist focus on **Computer Vision**

SKILLS

- Full Stack: **Python**, **JavaScript**, **Java**, **TypeScript**, **React**, **HTML**, **CSS**, **PostgreSQL**, **Kotlin**, **C++**, **Firebase**
- Tools/Libraries: **Unity (C#)**, **Git**, **Mercurial**, **Docker**, **Kubernetes**, **OpenCV**, **PyTorch**, **Pandas**, **JsPsych**, **bash scripting**